

Generate Sub-Arrays

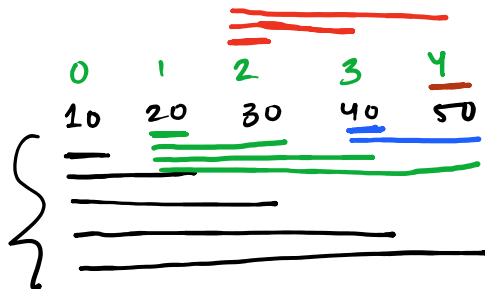
Given an array of **N** integers, design an algorithm to generate all of its **sub-arrays**.

Note : We define a **sub-array** of an array as a **contiguous** part of the given array.

Example

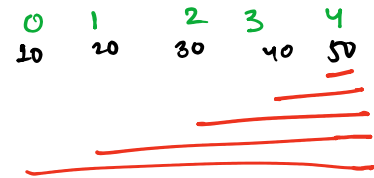
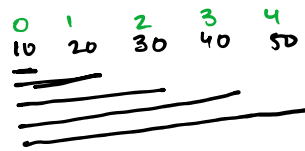
Input

0	1	2	3	4
10	20	30	40	50



$$\Rightarrow 0 \leq i \leq n-1 \quad i \leq j \leq n-1$$

i	j	# subarrays	$\left. \begin{array}{c} \sum \\ \sim 1^2 \\ = \end{array} \right\} = \frac{n(n+1)}{2}$
0	0 to $n-1$	n	
1	1 to $n-1$	$n-1$	
2	2 to $n-1$	$n-2$	
$\vdots$	$\vdots$	$\vdots$	
$\vdots$	$\vdots$	$\vdots$	
$n-1$	$n-1$ to $n-1$	1	



Maximum Subarray Sum

Given an array of **N** integers, design an algorithm to find the **maximum subarray sum**.

Example

Input

0	1	2	3	4	5	6	7	8
-2	1	-3	4	-1	2	1	-5	4

Output : 6

"udiksha"

"u"

"udik"

"ud"

"udiks"

"udi"

"udiksh"

"udiksha"

"a"

"iksha"

"ha"

"diksha"

"sha"

"udiksha"

"ksha"

Cumulative / Prefix Sum

0 -2 -1 -4 0 -1 1 2 -3 1

arr

	0	1	2	3	4	5	6	7	8
	-2	1	-3	4	-1	2	1	-5	4

csum

0	1	2	3	4	5	6	7	8	9
0	-2	-1	-4	0	-1	1	2	-3	1

psum

Handwritten annotations: A bracket above the array from index 0 to 4 is labeled $1 = 9$. A bracket below the array from index 0 to 4 is labeled $1 + 1 = 10$. A bracket below the array from index 0 to 9 is labeled $1 + 1 = 10$.

$i+1 = 10$

10 20 30 40 50

10 30 60 100 150

Prefixes

subarray that starts at 0th idx

Suffixes

subarrays that ends at $i-1$ th idx

$$psum_i = \text{sum}(arr[0 \dots i-1]) = psum_{i-1} + arr_{i-1}$$

$$psum_{i-1} = \text{sum}(arr[0 \dots i-2]) \quad i \geq 1$$

arr

$s_{ij} = psum[j+1] - psum[i]$

arr		0	1	2	3	4	5	6	7	8	
		-2	1	-3	4	-1	2	1	-5	4	
csum		0	1	2	3	4	5	6	7	8	9
		0	-2	-1	-4	0	-1	1	2	-3	1

Maximum Subarray Sum using Kadane's Algorithm

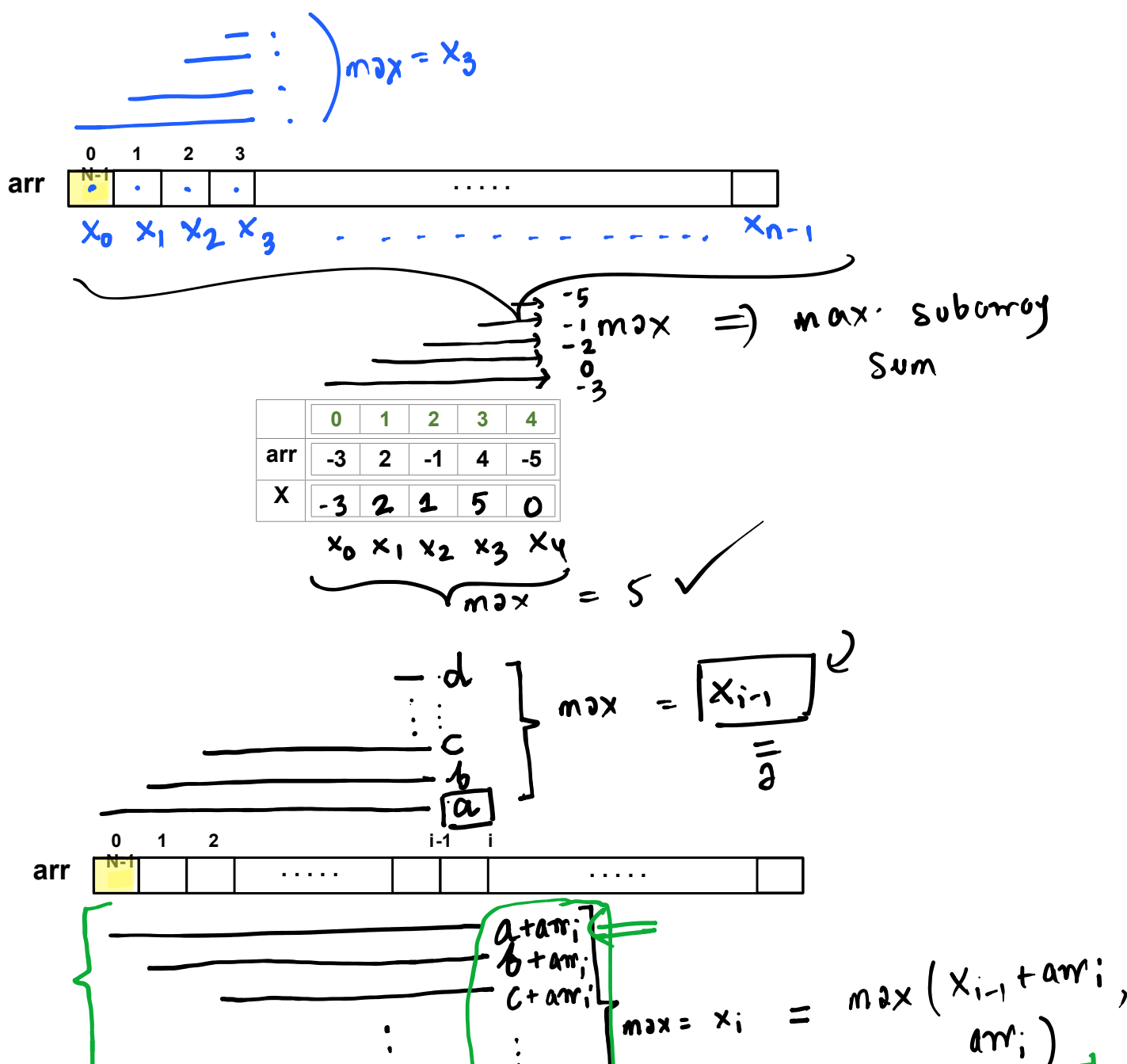
Given an array of **N** integers, design an algorithm to find the **maximum subarray sum**.

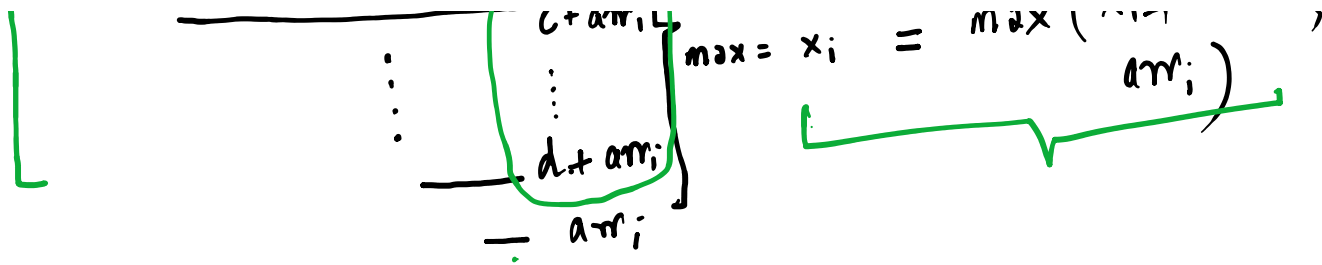
Example

Input

0	1	2	3	4
-3	2	-1	4	-5

Output : 5





→

	0	1	2	3	4
arr	-3	2	-1	4	-5
X	-3	2	1	5	0

$x_0 = \text{arr}_0$

$$x_i = \max(x_{i-1} + \text{arr}_i, \text{arr}_i)$$

$i \geq 1$

msf = ~~-3~~
2
5
=

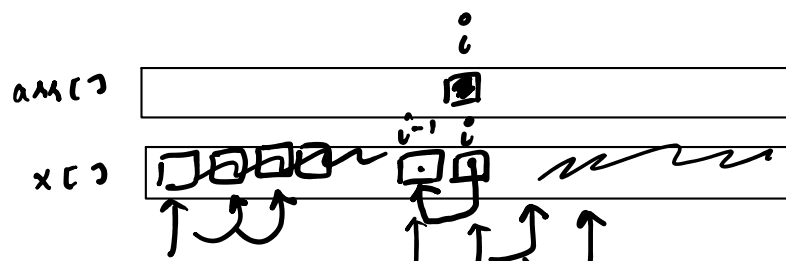
$$\begin{aligned} x_1 &= \max(x_0 + \text{arr}_1, \text{arr}_1) \\ &= \max(-3 + 2, 2) \\ &= \max(-1, 2) \\ &= 2 \end{aligned}$$

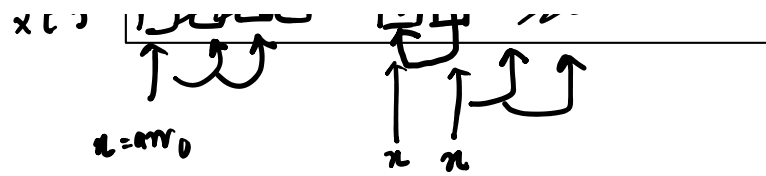
$$\begin{aligned} x_2 &= \max(x_1 + \text{arr}_2, \text{arr}_2) \\ &= \max(2 + (-1), -1) \\ &= \max(1, -1) \\ &= 1 \end{aligned}$$

$$\begin{aligned} x_3 &= \max(x_2 + \text{arr}_3, \text{arr}_3) \\ &= \max(1 + 4, 4) \\ &= 5 \end{aligned}$$

$$\begin{aligned} x_4 &= \max(x_3 + \text{arr}_4, \text{arr}_4) \\ &= \max(5 + (-5), -5) \\ &= 0 \end{aligned}$$

$$x_i = \max(x_{i-1} + \text{arr}_i, \text{arr}_i)$$





Maximum Circular Subarray Sum

Given an array of **N** integers, design an algorithm to find the **maximum circular subarray sum**.

Example

Input

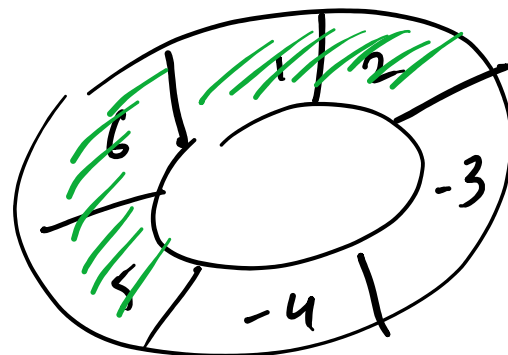
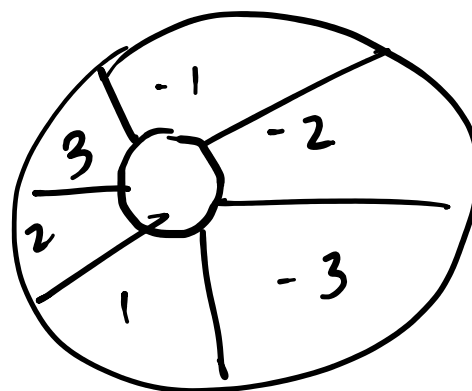
0	1	2	3	4	5
-1	-2	-3	1	2	3

Output : 6

Input

0	1	2	3	4	5
1	2	-3	-4	5	6

Output : 14



[HW] Maximum Product Subarray

<https://leetcode.com/problems/maximum-product-subarray/description/>

Generate Subsequences

Given a sequence of integers, design an algorithm to generate all of its **subsequences**.

Note : We define a **subsequence** of a given sequence as a sequence which can be obtained by deleting **zero** or **more** elements from the given sequence such that the **relative order** between the sequence elements **doesn't change**.

Example

Input



0	1	2
10	20	30



$\{10\}$

$\{10, 20\}$

$\{10, 20, 30\}$

$\{20\}$

$\{10, 30\}$

$\{\}$

$\{30\}$

$\{20, 30\}$

Dynamic Array

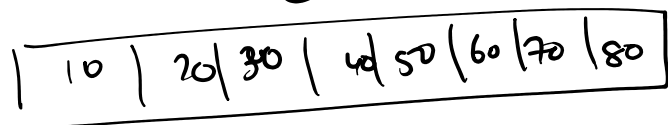
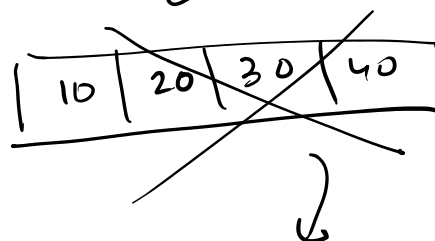
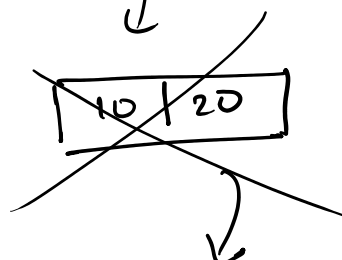
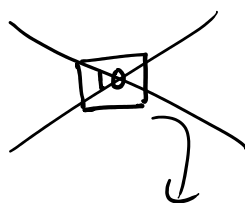
cap = 0
sz = 0

cap = 1
sz = 1

cap = 2
sz = 2

cap = 4
sz = 4

cap = 8
sz = 8



vector

#include <vector>

int x;
char ch;
bool b;

vector<int> v;
type name

```
v.push-back(10);  
cout << v.capacity(); // 1  
cout << v.size(); // 1  
v.push-back(20);  
cout << v.capacity(); // 2  
cout << v.size(); // 2
```

2001
⇒ out 22 v.size(); // 2
⋮